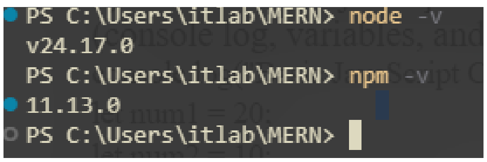
Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester IV)
MERN Practical

Practical – I

Roll No. : T048	Name: Lavkush Vishwakarma
Class : TYIT	Date/Time of Submission : 20/06/2026

Aim: Environment Setup and JavaScript Basics.

1.A). Install Node.js and Visual Studio Code, and verify installation.

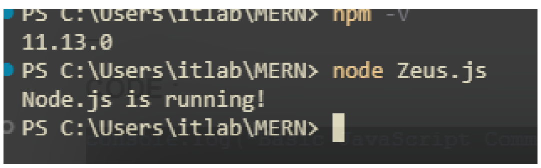


1.B). Create a simple Node.js file and run it using the terminal.

SOURCE CODE:-

```
console.log("Node.js is running!");
```

OUTPUT:-



1.C). Write a Node.js script demonstrating basic JavaScript commands (console log, variables, and arithmetic operations).

SOURCE CODE:-

```
console.log("Basic JavaScript Commands");
let n1 = 50;
let n2 = 60;
let sum = n1 + n2;
let difference = n1 - n2;
let product = n1 * n2;
let quotient = n1 / n2;
console.log("Number 1 =", n1);
console.log("Number 2 =", n2);
console.log("Addition =", sum);
console.log("Subtraction =", difference);
console.log("Multiplication =", product);
console.log("Division =", quotient);
```

OUTPUT:-

```
PS C:\Users\itlab\MERN> node Zeus.js
Basic JavaScript Commands
Number 1 = 50
Number 2 = 60
Addition = 110
Subtraction = -10
Multiplication = 3000
Division = 0.8333333333333334
```

1.D). Create and run a simple program using variables and functions.

SOURCE CODE:-

```
let n1 = 23;
let n2 = 19;
function add(n1, n2) {
   console.log("Addition =",n1+n2);
}
function sub(n1, n2) {
   console.log("Subtraction =",n1-n2);
}
add(n1, n2);
sub(n1,n2);
console.log("Lavkush")
```

OUTPUT:-

```
PS C:\Users\itlab\MERN> node zeus2.js
Addition = 42
Subtraction = 4
Lavkush
```

1.E). Demonstrate use of operators and control statements in JavaScript.

SOURCE CODE:-

```
let a = 10;
let b = 5;
console.log("Addition:", a + b);
console.log("Subtraction:", a - b);
console.log("Multiplication:", a * b);
console.log("Division:", a / b);
if (a > b) {
   console.log("a is greater than b");
} else {
   console.log("b is greater than a");
}
console.log("Lavkush")
```

OUTPUT:-

```
PS C:\Users\itlab\MERN> node zeus3.js
Addition: 15
Subtraction: 5
Multiplication: 50
Division: 2
a is greater than b
Lavkush
```

1.E.2).

SOURCE CODE:-

```
let day = 3;
switch (day) {
   case 1:
      console.log("Monday");
      break;
   case 2:
      console.log("Tuesday");
      break;
   case 3:
      console.log("Wednesday");
      break;
   default:
      console.log("Invalid Day");
}
console.log("Lavkush")
```

OUTPUT:-

```
PS C:\Users\itlab\MERN> node zeus4.js
Wednesday
Lavkush
```

1.B). Write programs using loops and template literals.

SOURCE CODE:-

```
let num =3;
console.log(`Multiplication Table Of ${num} `);
for(let i=1;i<=10;i++){
   console.log(`${num} x ${i} = ${num*i} `);
}
console.log("Lavkush")
```

OUTPUT:-

```
PS C:\Users\itlab\MERN> node zeus5.js
Multiplication Table Of 3
3 x 1 = 3
3 x 2 = 6
3 x 3 = 9
3 x 4 = 12
3 x 5 = 15
3 x 6 = 18
3 x 7 = 21
3 x 8 = 24
3 x 9 = 27
3 x 10 = 30
Lavkush
PS C:\Users\itlab\MERN>
```

1.C). A school wants to develop a simple application using Node.js. Create a Node.js application that calculates and displays a student's result by applying basic JavaScript concepts.

SOURCE CODE:-

```
let studentName = "Zeus";
let rollNumber = 101;
let subject1 = 85;
```

```
let subject2 = 78;
let subject3 = 92;
// Store marks in an array
let marks = [subject1, subject2, subject3];
// Step 4: Functions
function calculateTotal(markArray) {
   let total = 0;

   for (let mark of markArray) {
      total += mark;
   }
   return total;
}
function calculatePercentage(total, numberOfSubjects) {
   return total / numberOfSubjects;
}
// Call functions
let totalMarks = calculateTotal(marks);
let percentage = calculatePercentage(totalMarks, marks.length);
// Step 6: Grade using if-else
let grade;

if (percentage >= 90) {
   grade = "A+";
} else if (percentage >= 75) {
   grade = "A";
} else if (percentage >= 60) {
   grade = "B";
} else if (percentage >= 50) {
   grade = "C";
} else if (percentage >= 35) {
   grade = "D";
} else {
   grade = "Fail";
}
// Step 7: Display subject marks using loop
console.log("Subject Marks:");
for (let i = 0; i < marks.length; i++) {
   console.log(`Subject ${i + 1}: ${marks[i]}`);
}
// Step 8: Template Literals
console.log(`
-------------------------------
Student Result System
-------------------------------
Student Name : ${studentName}
Roll Number  : ${rollNumber}

Total Marks  : ${totalMarks}
Percentage : ${percentage}%
Grade        : ${grade}
```

`);
console.log("Lavkush")

OUTPUT:-

```
PS D:\Users\Lavkush\Study\MERN> node studentResult.js
Subject Marks:
Subject 1: 85
Subject 2: 78
Subject 3: 92

-------------------------------
Student Result System
-------------------------------
Student Name : Zeus
Roll Number  : 101

Total Marks  : 255
Percentage : 85%
Grade        : A

Lavkush
```